Underrepresentation in K-12 CS: A Systematic Literature Review

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***Abstract*—Over the last several decades, computers and computing technology have advanced to become ubiquitous in our society. It has permeated the lives of nearly every American, yet the people driving these innovations have remained relatively the same demographic. It is widely understood that women, Black, Brown, Indigenous, low-income, and rural people remain persistently underrepresented across the field of computer science in the US. This paper presents a systematic review of the literature to explore the current state of CS education in K-12 for these specific groups. Our methodology comprises a structured search and filtering of articles from the IEEE, ACM, and ERIC digital libraries. The resulting papers are then examined, categorized, quantified, and visualized. Through analysis, this review provides a comprehensive view of the current state of CS education in K-12 for underrepresented groups, providing insight and identifying areas of improvement and future interventions.**

***Index Terms*—Computer Science, Computer Science Education, K-12 Education, Underrepresented Groups**

I. Introduction

Within the last century, computer science (CS) has become a fundamental driver of innovation, economic growth, and social infrastructure. As the field expands, the demand for a skilled and diverse computing workforce grows exponentially. Because a bachelor's degree is the typical requirement for workers in computer and information technology, it is unsurprising that the field is classified as having a high barrier to entry [1] [2].

The current literature demonstrates the persistent underrepresentation of women, Black, Brown, and Indigenous populations across the technology sector in the United States. In the latest Bureau of Labor Statistics annual averages, women made up 27.5% of workers in computer and mathematical occupations, compared with 47.1% of the total employed U.S. workforce; Black workers accounted for 9.5% and Hispanic/Latino workers 8.7%, compared with 12.7% and 20.0% of the total workforce, respectively [3].

In addition, barriers continue to exclude students from low-income households and rural communities, limiting their access to the economic mobility that careers in computing can often provide [4]. This lack of diversity in the workplace can easily be traced back to disparities in K-12 education; thus, the most critical time to affect change and broaden participation in CS is in K-12 [5]. This paper reviews the literature on the current state of CS education in K-12 and discusses the approaches, pedagogies, and tools used by educators.

In the following chapters, we first present our methodology for finding studies relevant to our research questions. Then the results chapter comprises the current data and trends from the final selection of articles. Lastly, in the discussion chapter, we synthesize and visualize the current state of the literature to provide insight into the research question, RQ: *What is the current status of CS education for women, Black, Brown, Indigenous, rural, and low-income populations within K-12?* We attempt to answer this question by examining:

1) What grade levels are most targeted by the current body of research?
2) What languages do they learn?
3) What tools do they use?
4) Which groups are being the least examined?
5) In what ways are underrepresented groups supported in CS education?

II. Methodology

Our primary goal throughout this review was to encompass as many relevant publications on the topic as possible while still ensuring the integrity and quality of the documents. In every part of this process, extra caution is taken to ensure that researcher bias is kept to a minimum. To assist in this objective, inspiration and direction were taken directly from the guidelines presented by KCeele [6]. Reducing researcher bias begins with predefining the review protocol. The components of our review protocol are:

Review Protocol:

1) Development of the research questions

2) Selection of the libraries
3) Query Strategy
4) Selection Criteria
5) Data Extraction
6) Synthesis
7) Dissemination

A. Library Selection

The articles were selected from the following sources: ACM Digital Library, IEEE Xplore, and the Education Resources Information Center (ERIC). Both ACM and IEEE are well established and highly regarded organizations for CS education research. Thus, the majority of relevant papers were expected to be found in either of those libraries. The ERIC library was added to capture any articles that may be missing from ACM and IEEE libraries. ERIC comprises articles with an education perspective.

B. Query Strategy

To formulate effective queries, we first generated search terms based on our research question. Our goal with these terms was to encapsulate the focus of as many relevant studies as possible. To minimize the bias in the results, we limited ourselves to terms directly from our research questions and interchangeable terms (e.g., 'computer science education' and 'computing education'). As for ensuring scope, we utilized a three-tier approach. The primary terms pertain to the desired topic: computer science education. The secondary terms relate to the group of students we are interested in: K-12. The final tertiary terms relate to the specific group the study focused on: underrepresented groups (e.g. women, Black, Brown, Indigenous, low-income, and rural populations). Our final terms are as follows:

- Primary Terms: "computer science education", "computing education", "computational thinking", "programming", and "coding"
- Secondary Terms: "K-12", "K12", "primary school", "elementary", "middle school", "high school", and "secondary school"
- Tertiary Terms: "underrepresented", "women", "girls", "female", "black", "african american", "brown", "latin", "hispanic", "indigenous", "low-income," "rural," "Native American," "American Indian," and "Alaska Native"

The final queries were constructed using boolean logic to combine these terms into a single search string. Using these search strings alone, our search yielded well over 400 results across the libraries.

C. Selection Criteria

To refine the preliminary results into a more usable and effective pool of sources, the following selection criteria are established.

Selection Criteria

1) All articles should be full research articles.
2) Study participants should be K-12 students or K-12 educators in CS or a related field.
3) The study should be conducted in the US.
4) The study should be conducted between January 1st, 2010, and October 1st, 2025.
5) Any study before 2025 should be cited by at least three other studies.

After filtering based on the selection criteria, the final list resulted in 57 articles, providing an accurate representation of the current state of the literature.

D. Data Extraction

The following data was collected for each paper:

- Research Questions
- Location
- Participants (Size and Group)
- Grade
- Programming Language(s)
- Instructional Tool(s)
- Target Population
- Connections to existing Initiatives/Works

After the initial review and collection process, we revised our model to include categorical data such as "Study Type" and "Outcome Type". The selected literature was then reexamined and categorized.

III. Results

A. Dataset

A total of 57 articles were included in the final corpus following the screening and selection process. Their distribution across the three digital libraries is presented in Table I. As expected, the majority of publications originated from computing-focused venues (ACM and IEEE), reflecting the strong emphasis on computer science education research within these communities relative to education-focused repositories such as ERIC.

Temporal trends in publications are illustrated in Figure 1. Overall, the number of publications increases over time, with a noticeable rise after 2016.

The selected papers were further categorized in Table II into three primary study types: case studies, surveys, and reviews. Case studies represent the dominant methodology and include empirical investigations conducted within a specific classroom, school, or program setting. These studies are typically intervention-based and involve the direct implementation of curricula, tools, or pedagogical approaches. Survey papers

TABLE I
Number of Papers by Library

Library	Papers	Total	Percent
ACM	[4], [5], [7]–[30]	26	45.6%
IEEE	[31]–[48]; [49]	19	33.3%
ERIC	[50]–[61]	12	21.1%

TABLE II
NUMBER OF PAPERS BY STUDY TYPE

Study Type	Papers	Total	Percent
Case Study	[5], [7], [8], [10]–[13], [15], [17]–[19], [22]–[24], [27], [30], [32]–[36], [38], [40]–[46], [48]–[52], [55]–[57], [59], [60]	39	68.4%
Survey	[4], [9], [16], [20], [21], [25], [26], [29], [37], [54]	10	17.5%
Review	[14], [28], [31], [39], [47], [53], [58], [61]	8	14.0%

solely discuss the results and implications of a survey conducted in a specific group. Review papers can be systematic literature reviews, and/or meta-analyses articles that synthesize findings from pre-existing data.

Figure 2 demonstrates the geographic distribution of the studies for those studies that clarify their location. Not represented in the figure are the four studies in the selected literature that were conducted on a national level. The distribution reveals that the majority of the US states did not perform any studies, and half of the studies are concentrated in only three states, California, Illinois and New York.

B. Research Questions

What is the current status of CS education for the underrepresented populations?

The overarching goal of this review is to assess the current state of K–12 computer science education for underrepresented populations in the United States. Given the scope of this objective, the analysis is structured around five more specific research questions. The following sections present the results corresponding to each research question.

1) RQ #1: What grade levels are most targeted by the current body of research?

Figure 3 shows the distribution of studies across grade levels. The results indicate that the literature spans across the whole range from K-12. However, the main body of work is concentrated primarily middle school through high school grades, which is consistent with other population groups.

2) RQ #2: What languages do they learn?

Before proceeding with questions 2 and 3, the following distinction must be made between languages and tools. Programming languages refer specifically to the formal languages used to write code (e.g., Python, Java, Scratch). In contrast, tools encompass the broader collection of software, environments, or hardware used to facilitate learning. This includes robot kits, circuits, and other domain specific tools.

Figure 4 shows the distribution of programming languages used across relevant studies (note that some papers either did not utilize a programming language or specify the one used). It is evident that there is a high preference for the Scratch programming language. Out of all the papers that specified a programming language, 50% of them utilized Scratch.

When examining the data with respect to grade level demonstrated by Figure 5, Scratch and Alice, the two block-based languages, seem to dominate the lower grade levels. However, for the 9th grade and higher, Python becomes the prevailing programming language.

Overall this progression suggests that block-based environments are often used in earlier learning ages while text-based languages become more common as a student's age increases.

3) RQ #3: What tools do they use?

Figure 6 illustrates the distribution of tools used across the reviewed studies. Scratch-based environments, robotics platforms (e.g., LEGO, Arduino), and educational platforms such as Code.org represent the most commonly utilized tools. These tools collectively account for a substantial proportion of the interventions documented in the literature.

The tools-by-grade heatmap further highlights how tool usage varies across grade levels, Fig. 7. Block-based tools such as Scratch and Kodable are concentrated in elementary and middle school grades, while robotics platforms and development tools such as Arduino, Unity, and MIT App Inventors appear more frequently in middle and high school contexts.

Additionally, several tools such as e-textiles and EarSketch are used across multiple grade levels, often in contexts designed to increase engagement through creative or culturally relevant applications. Overall, the results indicate a diverse

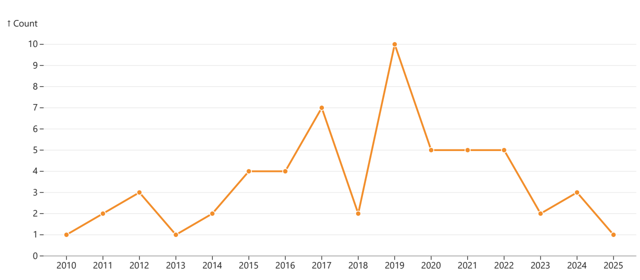

Fig. 1. Timeline of Papers

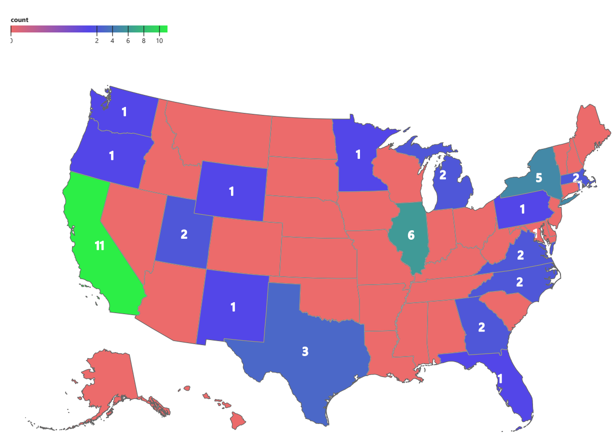

Fig. 2. Distribution of Papers by Location

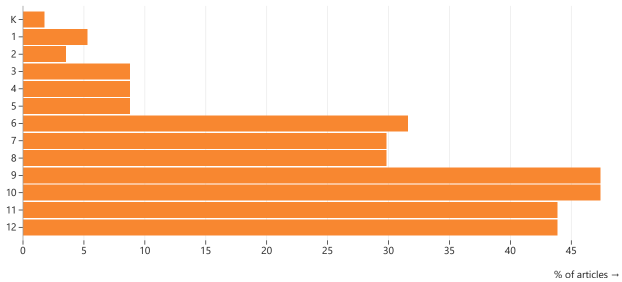

Fig. 3. Distribution of Papers by Grade

variety of tools, with a strong emphasis on interactive and project-based learning environments.

4) RQ #4: Which groups are being the least examined?

To address this question, each paper is categorized by its target population. For studies with multiple underrepresented groups, the following distinctions were made:

- Studies exclusively focused on three or less groups would be counted for each of those groups. For example, a study focused on Black and Latina women would be classified under Black, Latinx, and women categories.
- Studies that involved all the underrepresented groups as a single group or examined data for many groups were classified as "unspecified".

Figure 8 presents the distribution of studies by target population group. The results indicate that women/girls and broadly defined underrepresented groups account for the largest share of studies, while groups such as rural populations are less frequently represented .

Table III provides further insight into study scale across populations. Low-income populations have the highest median sample size (105), suggesting that while fewer studies may target this group, they are often conducted at larger scales. In contrast, rural populations have the smallest median sample size (34), indicating both lower representation and smaller study scope. Black and Latinx populations fall in the mid-range (40–41), and studies focusing on women and unspecified groups tend to involve larger samples (60).

These findings suggest an imbalance in the literature, where women received considerably more attention than other groups. Certain groups like rural populations, receive much less attention in both frequency and scale of study. Moreover, one group is omitted altogether from the data – Native Populations.

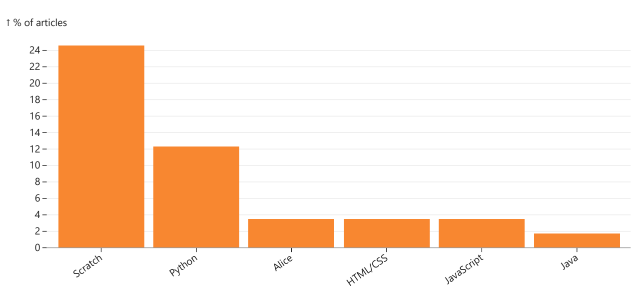

Fig. 4. Distribution of Papers by Language

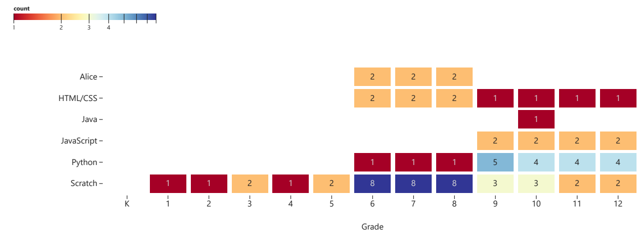

Fig. 5. Languages by Grade Heatmap

5) RQ #5: In what ways are underrepresented groups supported in CS education?

To examine how underrepresented groups are supported within CS education, studies were categorized into three outcome types: access, engagement, and learning.

- *Access* refers to studies aimed at increasing exposure or participation in CS, such as outreach programs.
- *Engagement* studies focus on how to sustain student interest and participation over time.
- *Learning* pertains to measurable educational outcomes and performance metrics across the groups. These studies often aim to explore ways to achieve equity in learning outcomes.

To maintain consistency, papers were classified under the single outcome type that best categorized the study. The distribution of studies by outcome type is presented in Table IV. Learning-focused studies constitute the largest proportion of the literature (40.4%), followed by access-oriented studies (36.8%), while engagement-focused work represents the smallest share (22.8%). This distribution indicates a moderate preference towards learning and access focused studies. However, when considering their proportions over recent years, there seems to be a clear imbalance with engagement-focused studies being even less represented.

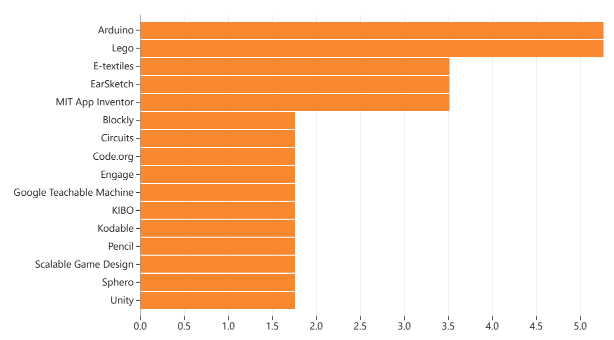

Fig. 6. Distribution of Papers by Tool

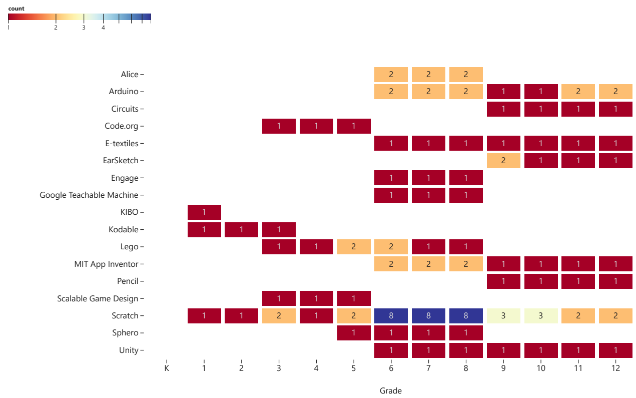

Fig. 7. Tools by Grade Heatmap

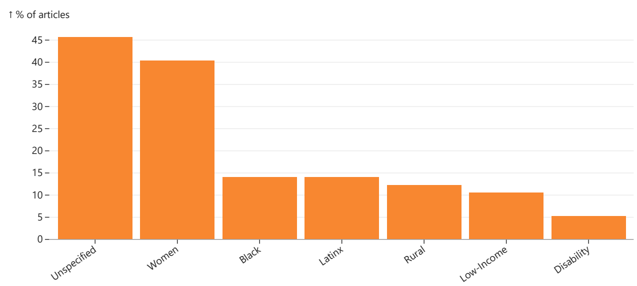

Fig. 8. Distribution of filtered papers by population group.

Temporal trends in outcome type are illustrated in Figure 9. In more recent years, there has been a noticeable increase in learning-focused studies, indicating a shift toward evaluating student outcomes and equity in achievement. When only considering literature within the last 6 years, learning-focused studies represent the majority of papers, and engagement-

TABLE III
MEDIAN SAMPLE SIZE BY POPULATION GROUP

Group	Number of Case Studies	Median Sample Size
Unspecified	17	60
Women	15	60
Black	6	40
Latinx	6	41
Low-Income	4	105
Rural	4	34
Overall		50

The median was calculated with case study papers only.

TABLE IV
NUMBER OF PAPERS BY OUTCOME TYPE

Outcome Type	Number of Papers	Percentage
Access	21	36.8%
Engagement	13	22.8%
Learning	23	40.4%
Total	57	

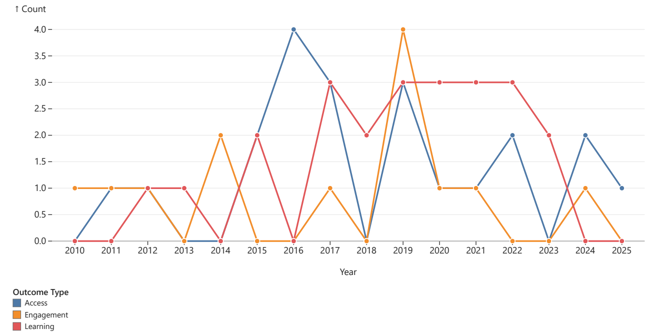

Fig. 9. Timeline of Papers by Outcome Type.

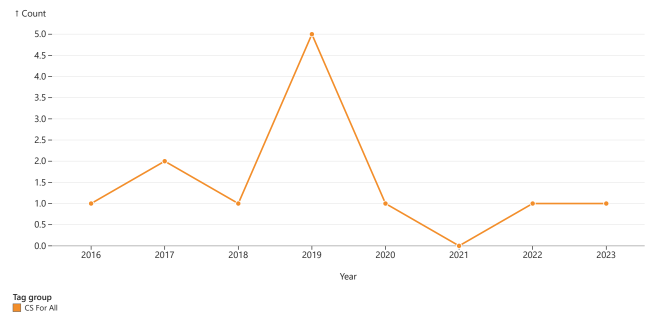

Fig. 10. Timeline of Papers by Connection to CS For All Initiative.

focused studies only represent 14% of the literature.

Overall, these findings display a lack of literature on engagement within the last few years. Instead, the literature has shifted more towards achieving equity in learning outcomes. This comparatively lower representation of engagement-focused studies suggests that sustaining long-term interest and participation among underrepresented groups remains a less explored area within current research.

IV. Discussion

The findings presented in this review thoroughly examine several factors of CS education. Through our search methodology, the data compiled in this review spans a wide scope of papers, making it very representative of the current CS education literature. However, it is not only the data that the paper analyzed that is important, but also the data not present in this review. Among all the selected literature, one population group was completely absent from the literature: Native American and Indigenous students. Even throughout the studies with unspecified underrepresented groups, Indigenous students remain missing from the conversation. Such an absence points to a serious gap in the literature, especially given the stated goal of broadening participation across underrepresented groups. Although it might be possible that literature exists outside the reach of the search process, its disappearance from the selected journals indicates that Indigenous students are not being adequately represented in the most visible research spaces. Additionally, population imbalances are made evident in this review. Research on female students comprises an overwhelmingly large portion of the literature. In contrast, rural populations appear much less often, with rural-focused studies generally being smaller in scale (smaller median sample sizes).

A. Publication Trends and CS For All

One clear pattern is the increase in research activity in recent years, particularly after 2016. Much of this growth appears to coincide with the launch of the CS for All initiative [62]. Figure 10 displays the output of CS For All connected papers over time. Compared to the timeline of total papers shown by Figure 1, the CS For All initiative's impact on the literature is apparent. This suggests that national efforts to broaden participation in computing positively contribute to both the development of new interventions and the increase in research for these groups.

B. Outcome Emphasis

Regarding outcomes, the literature appears to emphasize access and learning more often than engagement. Access and learning account for much of the reviewed work, whereas engagement appears less frequently, particularly in more recent years. This distinction is important because access, engagement, and learning represent different but related forms of support. The literature currently provides substantial evidence on access and increasing evidence on learning, but it provides less evidence on the factors that help students remain involved over time. This gap is especially important for underrepresented groups, for whom persistence and belonging may be as critical as initial participation. [63]

C. Overarching Studies with National Scope

Beyond localized interventions and case studies, there were four survey papers with a national scope. These large-scale papers help frame the broader landscape of underrepresentation in K–12 CS education. Wang et al. (2016) [4] show that, at a national level, many students, parents, teachers, and administrators still hold incomplete or stereotypical views of computer science. This only worsens the issue of uneven access to CS opportunities across gender, race, and income. An earlier national survey study focused on gender (Wang et al. 2015 [16]) further found that encouragement and exposure are among the strongest controllable factors influencing girls' decisions to pursue computing. Other especially important factors are family support, role models, and perceptions of CS careers.

Shifting from student perceptions to instruction, Banilower and Craven [20] show that stronger classroom experiences are associated with teacher preparedness, substantial professional development, and the coherence of instructional materials. Lastly, Sax et al. [9] find that AP CS Principles (CSP) appear to broaden participation for women, Black, Latinx, lower-income, and first-generation students into advanced CS coursework. However, AP CSP alone is not enough to function as a pipeline into a CS career. The class only increases access, but retaining the engagement of these students after the class with additional courses provides a much better pipeline.

Together, these papers describe underrepresentation in CS as more than just an access problem. Instead, it requires a larger emphasis on long-term engagement and factors that may affect retention. Both Sax et al. and Wang et al. (2015) corroborate the importance of engagement, which has been lacking as a focus in the current literature, as described in the previous section.

D. Culturally Relevant Pedagogy

Culturally Relevant Pedagogy is a popular pedagogical framework that utilizes students' backgrounds, culture, and experiences to improve learning outcomes [64]. The approach relies on three criteria:

- Students must experience academic success
- Students must develop and/or maintain cultural competence
- Students must develop a critical consciousness through which they challenge the status quo of the current social order

Despite its age, this pedagogical model is still very relevant in the current literature. Culturally Relevant Pedagogy was utilized in several studies throughout the selected papers and referenced in even more. Madkins et al. [32] study three separate programs to identify three recurring themes across them: engagement and relevance, confidence and identity, and social justice. Across those cases, students showed increased perceptions of the usefulness of CS, improved clarity about the field, stronger belief in the cultural relevance of CS, and stronger self-regulation and academic outcomes. Ryoo [5] reports a closely related pattern in formal ECS classrooms, finding that engagement was strongest when teachers demystified CS by connecting it to everyday life, relevant social issues, communities, and student voices and perspectives. Additionally, case studies such as Franklin et al. [11] utilize the framework to guide outreach programs. The paper finds that a summer camp built around culturally relevant themes could support not only interest but also achieve positive measurable learning outcomes of core CS concepts. Collectively, these three papers demonstrate how culturally relevant approaches in CS support more than initial access alone; they also strengthen engagement, identity, and learning.

Thus, Culturally Relevant Pedagogy can become a vital framework to combat the issues of engagement detailed in the previous sections. From the reviewed literature, the framework has consistently shown positive results in improving learning outcomes and increasing factors of engagement. Factors like perception of CS, retained interest, and stereotypes, which were outlined as important by the national survey studies [4], [9], [16], seem to be addressed directly through culturally relevant approaches. The framework accomplishes this as it connects CS to students' lives, communities, and identities. In this sense, culturally relevant pedagogy appears to be one of the more promising trends in the current literature, as it supports not only access to CS but also the deeper sense of relevance, confidence, and belonging that are crucial for long-term participation. While the existing evidence is still largely based on localized case studies, the consistency of the positive outcomes across different contexts suggests that this framework deserves greater attention in future research and broader implementation in K-12 CS education.

V. Conclusion

This systematic literature review examined the current state of K-12 computer science education for underrepresented populations in the United States through an analysis of 57 papers from ACM, IEEE, and ERIC. Across the selected literature, the findings show that research in this area has grown substantially over time, especially after 2016, coinciding with the launch of the National CS For All initiative.

The reviewed studies indicate that K-12 CS education research for underrepresented groups is concentrated most heavily in middle and high school contexts. In terms of instructional approaches across the grade bands, there appears to be a strong emphasis on interactive, creative, and project-based learning environments. This is reflected by the tools utilized, including Scratch, robotics platforms, e-textiles, EarSketch, and other educational technologies.

The results also reveal substantial disparities in the populations represented in the literature. A large portion of the reviewed work was comprised of studies focused on women and underrepresented groups as a whole. Furthermore, Native American and Indigenous students were absent from the selected literature altogether, indicating a meaningful imbalance in visible research coverage. Geographic representation is also skewed, with the majority of studies residing in only a few states, particularly California, Illinois, and New York.

Regarding outcome types, the literature places the greatest emphasis on learning and access, while engagement receives comparatively less attention. This distinction is important because access, engagement, and learning represent different dimensions of participation in CS education. The literature also suggests that underrepresentation is shaped not only by access to coursework but also by perceptions of CS, teacher preparation, instructional quality, and sustained participation.

Finally, the review highlights culturally relevant pedagogy as one of the more prominent and promising approaches represented in the literature. Across several studies, culturally relevant approaches were associated with increased CS engagement, stronger identity development, improved perceptions of CS, and measurable learning outcomes like test results.

VI. Future Work

The findings from this review point to several clear directions for future research in K-12 computer science education. First, future research attention should be given to Native American students, as that population was entirely absent from the reviewed work. Additionally, expanding access for rural populations should be further explored, particularly in the various states that were completely unrepresented in this review.

Second, future studies should move beyond increasing access and place more focus on improving long-term engagement. As demonstrated by some of the long-term, large-scale studies, initial participation does not necessarily ensure that students remain involved in CS over time. Future studies should therefore examine the factors that support long-term interest, identity development, and retention among underrepresented students.

Third, more large-scale and longitudinal studies are needed. Much of the reviewed literature consisted of short-term case studies or localized interventions. Although these studies provide valuable insight into specific programs and contexts, they are limited in their ability to explain long-term outcomes or broader patterns across schools, states, and student populations.

Lastly, future work should continue examining the role of national initiatives such as CS For All in shaping both research activity and classroom practice. The increase in CS education research after 2016 suggests that broad policy and funding initiatives can heavily influence the development of interventions and studies on underrepresented groups.

VII. Limitations

There are some limitations present in the study. First, this study is limited to research conducted in the United States and cannot necessarily be generalized to the context of other nations. The results are also limited by the search strategy and database selection.

Next, the citation-based inclusion criterion may have affected the representation of newer studies. Since we only selected studies before 2025 with at least three citations, some recent or emerging work may have been excluded before it had enough time to accumulate citations. This may partly explain why only a handful of papers from the last three years appeared in the final selection. However, research bias was mitigated as much as possible in this process as the criterion was selected before the selection process and not selectively done after the search.

Finally, the analysis is limited by the information reported in the selected papers. Some studies did not clearly specify details such as grade level, location, programming language, tools used, or participant demographics. As a result, certain trends in the visualizations and tables reflect only the information available from the reviewed articles. Despite these limitations, we believe this review provides useful insight into the current status of CS education for underrepresented populations that can inform future efforts to broaden participation in CS.

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